

Interferometric Study of the Normal and Anomalous Zeeman Effects in Cadmium

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Abstract—The Zeeman effect corresponds to the lifting of atomic energy-level degeneracies in the presence of an external magnetic field. In this experiment, we studied the normal and anomalous Zeeman effects in cadmium emission lines using a Fabry–Pérot interferometer. The interferometric ring splitting was analyzed as a function of magnetic field in order to estimate the wavelength shift induced by the Zeeman interaction. We also investigated the polarization properties of the emitted lines for observation directions parallel and perpendicular to the magnetic field. From the normal Zeeman effect, we obtained an experimental estimate of the Bohr magneton. The full data set yielded $\mu_B = 15.14 \times 10^{-24} \text{ J T}^{-1}$, while restricting the analysis to the highest-field measurements, where the ring splitting was more reliably resolved, gave $\mu_B = 11.10 \times 10^{-24} \text{ J T}^{-1}$. These values are compared with the accepted value $\mu_B = 9.274 \times 10^{-24} \text{ J T}^{-1}$. The remaining discrepancy is attributed mainly to ring-identification uncertainty, optical misalignment, limited spectral splitting at low magnetic fields, and image-processing limitations.

I. INTRODUCTION

When an atom is placed in an external magnetic field, the interaction between the magnetic moments of the electrons and the applied field lifts the degeneracy of atomic energy levels. This phenomenon, first observed by Pieter Zeeman in 1896, provides direct evidence of the quantization of angular momentum and remains a central experiment in atomic spectroscopy [1]–[3].

In the simplest case, known as the normal Zeeman effect, the total spin is zero and the spectral line splits into three components. In the anomalous Zeeman effect, spin-orbit coupling and nonzero spin lead to more complex splitting patterns governed by the Landé g -factors of the initial and final states. The polarization of the emitted light also depends on the selection rule $\Delta M_J = 0, \pm 1$, giving rise to the so-called π and $\sigma^\pm$ transitions.

The wavelength shifts produced by the Zeeman effect are very small, typically on the order of 10^{-11} m , and cannot be resolved with a simple spectrometer. A Fabry–Pérot interferometer is therefore used to convert small wavelength differences into measurable changes in the radius of interference rings [4], [5].

The goals of this experiment were:

- to observe the normal and anomalous Zeeman effects in cadmium;
- to study the polarization of the split spectral components;
- to calibrate the magnetic field generated by the coils;
- to estimate the Bohr magneton from the measured spectral splitting;

- to identify the main experimental limitations of the measurement.

II. ELEMENTS OF THEORY

A. Angular Momentum Quantum Numbers

Atomic levels are described using the spectroscopic notation

$$^{2S+1}L_J,$$

where L is the orbital angular momentum quantum number, S the total spin quantum number, and J the total angular momentum quantum number. The magnetic quantum number M_J takes the values

$$M_J \in \{-J, -J+1, \dots, J-1, J\}.$$

In the absence of a magnetic field, states with different M_J are degenerate. The Zeeman effect lifts this degeneracy.

B. Zeeman Hamiltonian

In the absence of an external magnetic field, the atomic Hamiltonian can be written schematically as

$$\hat{H}_0 = \hat{H}_{\text{electrostatic}} + \hat{H}_{\text{spin-orbit}}. \quad (1)$$

In the presence of a magnetic field $\mathbf{B} = B\mathbf{e}_z$, an additional Zeeman interaction appears:

$$\hat{H} = \hat{H}_0 + \hat{V}_Z. \quad (2)$$

In the weak-field regime, where the magnetic interaction is smaller than the spin-orbit coupling, the energy shift of a level J, M_J is given by [2], [3]

$$\Delta E = \mu_B g_J M_J B, \quad (3)$$

where μ_B is the Bohr magneton and g_J is the Landé factor. The Bohr magneton is defined as

$$\mu_B = \frac{e\hbar}{2m_e} = 9.274 \times 10^{-24} \text{ J T}^{-1}, \quad (4)$$

where e is the elementary charge and m_e the electron mass [7]. The Landé factor in LS coupling is

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \quad (5)$$

For a transition between an upper level $(J', M_{J'})$ and a lower level (J, M_J) , the frequency shift is

$$h\Delta\nu = \mu_B B (g_{J'} M_{J'} - g_J M_J). \quad (6)$$

C. Normal and Anomalous Zeeman Effects in Cadmium

In the normal Zeeman effect studied here, we considered the cadmium transition

$$^1D_2 \rightarrow ^1P_1,$$

for which $S = 0$. In this case, $g_J = 1$ for both levels and the spectral line splits into three components corresponding to $\Delta M_J = 0, \pm 1$.

The anomalous Zeeman effect was studied using the transition

$$^3S_1 \rightarrow ^3P_2.$$

For this transition, the spin is nonzero and the Landé factors of the two levels are different. The splitting pattern is therefore more complex, and more than three components can be observed.

A schematic representation of the allowed transitions is shown in [Figure 1](#).

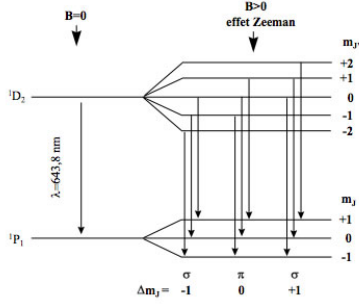


Figure 2: Effet Zeeman sur la transition $^1D_2 \rightarrow ^1P_1$ (raie rouge du Cadmium à 643,8 nm)

Figure 1: Schematic illustration of the allowed Zeeman transitions. The π transition corresponds to $\Delta M_J = 0$, while the $\sigma^\pm$ transitions correspond to $\Delta M_J = \pm 1$.

D. Polarization Selection Rules

The polarization of the emitted light is determined by the selection rule on M_J . The $\Delta M_J = 0$ transition is called the π transition and is linearly polarized parallel to the magnetic field. The $\Delta M_J = \pm 1$ transitions are called σ^+ and σ^- transitions.

When observing perpendicular to magnetic field, both π and σ components can be observed, with orthogonal linear polarizations. When observing along magnetic field direction, the π component is not observed, while the $\sigma^\pm$ components appear as circularly polarized light [2].

E. Fabry–Pérot Interferometry

A Fabry–Pérot interferometer produces circular interference fringes whose radius depend on the wavelength. A small Zeeman-induced shift $\Delta\lambda$ therefore leads to a splitting of the rings. By measuring the radius of the central, inner, and outer rings, the wavelength shift can be extracted [4], [5].

Let R_λ denote the radius of the unshifted ring, and $R_{\lambda 1}$, $R_{\lambda 2}$ the radius of the split rings. To first order, we obtain

$$\frac{R_\lambda^2}{R_{\lambda 2}^2 - R_{\lambda 1}^2} \approx \frac{\lambda^2}{4d\Delta\lambda}(p - 1 + \epsilon), \quad (7)$$

where d is the mirror spacing, p is the interference order, and ϵ is a constant offset related to the fractional interference order.

By plotting

$$\frac{R_\lambda^2}{R_{\lambda 2}^2 - R_{\lambda 1}^2}$$

as a function of p , the slope gives access to $\Delta\lambda$.

For the normal Zeeman effect, $\Delta\lambda$ is related to μ_B through

$$\Delta\lambda = \mu_B \frac{\lambda^2}{hc} B. \quad (8)$$

III. EXPERIMENTAL SETUP

A. Optical Setup

The experiment used a cadmium lamp placed at the center of an electromagnet. The current was limited to 10 A to avoid damaging the coils. The optical path consisted of a diaphragm, a Fabry–Pérot interferometer, a lens of focal length $f = 10$ cm, and a camera.

The optical setup is shown in [Figure 2a](#). The cadmium lamp emitted spectral lines whose Zeeman splitting was analyzed through the Fabry–Pérot interference rings.

B. Magnetic Field Calibration

Before analyzing the spectral splitting, the magnetic field generated by the coils was calibrated as a function of current using a Gaussmeter. The measured relation $B(I)$ is shown in [Figure 2b](#). This calibration was then used to associate each current value with a magnetic field amplitude.

IV. STUDY OF THE NORMAL ZEEMAN EFFECT

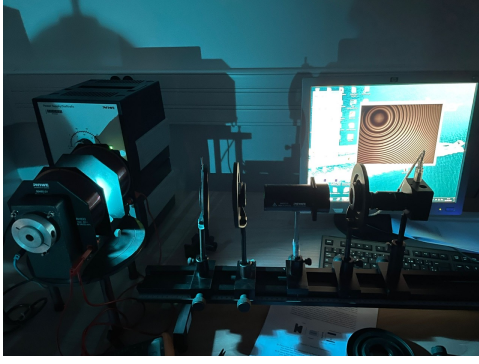
A. Polarization Observations

We first observed the polarization dependence of the split components. When the magnetic field was perpendicular to the observation direction, the π transition was linearly polarized parallel to the magnetic field, while the σ transitions were linearly polarized perpendicular to it. By rotating the polarizer, different components could therefore be enhanced or suppressed.

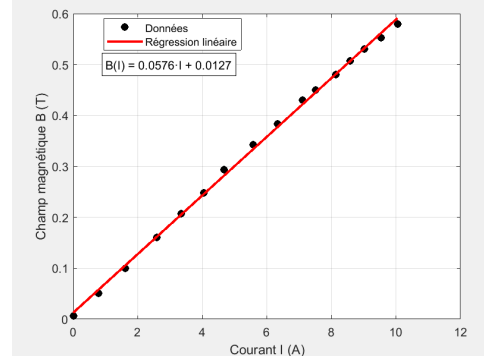
In [Figure 3](#), the ring pattern is shown for several polarizer orientations and geometries. In the perpendicular configuration, the outer rings corresponding to the σ transitions can be strongly attenuated for a suitable polarizer orientation. In the parallel configuration, the π component should ideally vanish, while the $\sigma^\pm$ components remain visible. The residual central intensity observed experimentally is likely due to imperfect alignment between the optical axis and the magnetic-field direction.

B. Ring Analysis Method

To estimate the Bohr magneton, we measured the splitting of the Fabry–Pérot rings for several values of the magnetic field. The images were processed using MATLAB to extract the intensity profile as a function of radial distance from the ring center. The radius R_λ , $R_{\lambda 1}$, and $R_{\lambda 2}$ were then estimated from the positions of the intensity maxima.

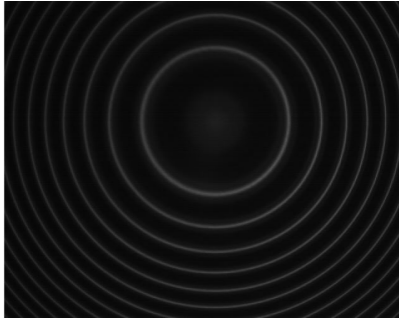


(a) Experimental optical setup.

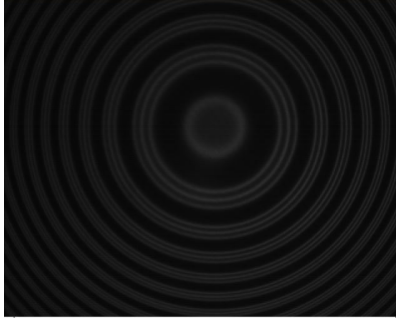


(b) Calibration of the magnetic field as a function of coil current.

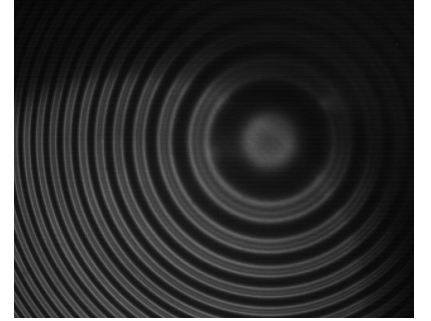
Figure 2: Experimental setup and magnetic-field calibration.



(a) Perpendicular geometry, polarizer angle 0° , $I = 8.06$ A.



(b) Perpendicular geometry, polarizer angle 10° , $I = 8.06$ A.



(c) Parallel geometry, arbitrary polarizer angle, $I = 7.78$ A.

Figure 3: Observation of the normal Zeeman-effect ring patterns under different polarization conditions.

At high magnetic fields, the splitting was sufficiently large to identify the inner and outer rings clearly. At lower magnetic fields, the splitting became too small compared with the linewidth and image noise, making the extraction unreliable. This limitation is illustrated in Figure 4.

C. Experimental Estimation of the Bohr Magneton

For each magnetic field, the wavelength shift $\Delta\lambda$ was extracted using Equation (7). The resulting values of $\Delta\lambda$ were then plotted as a function of B . According to Equation (8), the slope is proportional to μ_B .

In the analysis, we used the cadmium red-line wavelength

$$\lambda = 622 \text{ nm.}$$

Using the complete set of measured points, we obtained

$$\mu_B = 15.14 \times 10^{-24} \text{ J T}^{-1}. \quad (9)$$

This corresponds to a relative deviation of approximately 63% from the accepted value in Equation (4).

However, the low-field measurements were the least reliable because the ring splitting was only weakly resolved. Restricting the regression to the highest-field measurements,

$$I = 8.59 \text{ A, } 9.08 \text{ A, } 9.57 \text{ A, } 10.1 \text{ A,}$$

gave

$$\mu_B = 11.10 \times 10^{-24} \text{ J T}^{-1}, \quad (10)$$

corresponding to a relative deviation of approximately 19.7%. This improved agreement confirms that the main limitation was the ability to resolve the ring splitting accurately.

V. STUDY OF THE ANOMALOUS ZEEMAN EFFECT

A. Expected Splitting Pattern

For the anomalous Zeeman effect, we studied the cadmium green line associated with the transition

$$^3S_1 \rightarrow ^3P_2.$$

For the lower level $J = 1$, the magnetic quantum number can take three values:

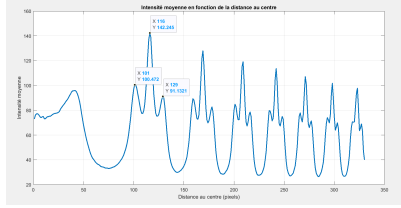
$$M_J \in \{-1, 0, 1\}.$$

For the upper level $J' = 2$, it can take five values:

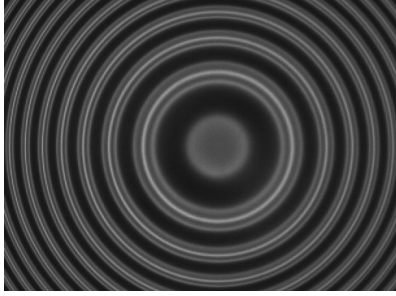
$$M_{J'} \in \{-2, -1, 0, 1, 2\}.$$

There are therefore 15 possible pairs of sublevels, but only those satisfying the dipole selection rule

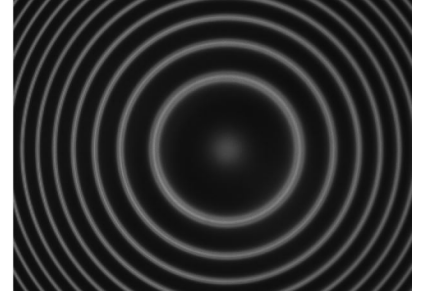
$$\Delta M_J = 0, \pm 1$$



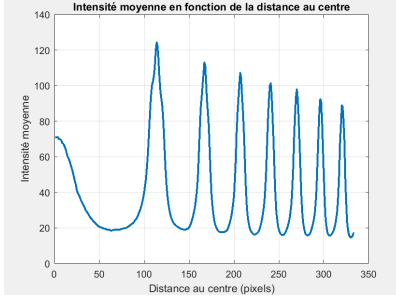
(a) Extraction of ring radius from the intensity profile for $I = 9.57$ A.



(b) Observed rings for $I = 9.57$ A.



(c) Observed rings for $I = 4.11$ A, where the splitting is difficult to resolve.



(d) Image-analysis output for $I = 4.11$ A, showing that reliable extraction of $R_{\lambda 1}$ and $R_{\lambda 2}$ is not possible.

Figure 4: Ring image analysis at high and low magnetic fields.

are allowed. This leads to nine allowed transitions.

Because the two levels have different Landé g -factors, the corresponding spectral shifts are not equally grouped into only three components, unlike in the normal Zeeman effect. The observed ring pattern is therefore more complex.

B. Experimental Observations

The anomalous Zeeman-effect rings are shown in Figure 6. The larger number of allowed transitions leads to a multiplicity of rings, which makes the identification of individual components difficult. In addition, at strong fields, rings from different interference orders begin to overlap.

The polarization behavior also differs from the normal Zeeman effect. Instead of isolating individual components, the polarizer tends to suppress or enhance groups of transitions sharing the same polarization character.

VI. DISCUSSION

A. Accuracy of the Bohr Magnetron Measurement

The measurement of the Bohr magneton was limited by several sources of uncertainty. The most important limitation was the extraction of the ring radius. At low magnetic fields, the spectral splitting was too small to clearly distinguish the inner and outer rings, while at high fields, the number of visible rings decreased and the patterns became sensitive to alignment and contrast.

An example of unreliable ring extraction is shown in Figure 7. Beyond a certain interference order, the inner and outer radius could no longer be measured reliably.

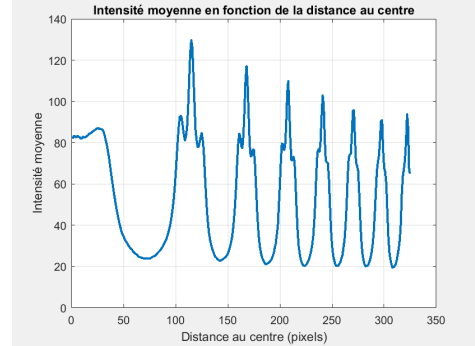
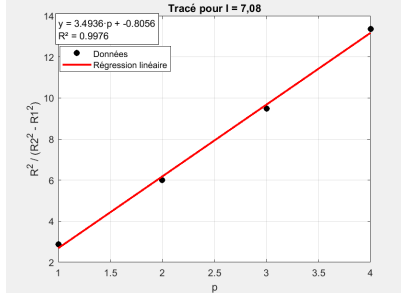


Figure 7: Example of image-analysis limitation for $I = 7.08$ A. Beyond $p = 5$, the radius $R_{\lambda 1}$ and $R_{\lambda 2}$ could not be extracted reliably.

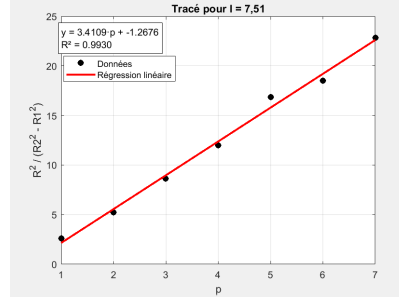
A second limitation was optical alignment. Some images showed asymmetry in the rings, likely due to imperfect centering of the optical bench, lamp, coil, Fabry-Pérot interferometer, lens, and camera. Because the radius-extraction method averaged over angular sectors, such asymmetry could bias the fitted radius.

Possible improvements would include:

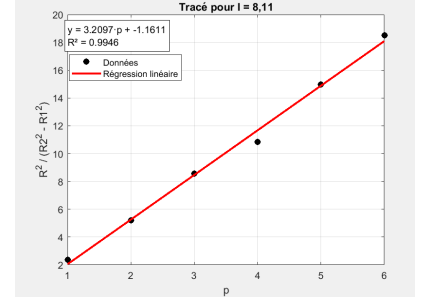
- improving the alignment of the optical bench and magnetic-field axis;
- optimizing camera exposure, gain, and contrast;
- fitting the rings using a two-dimensional circular or elliptical model;



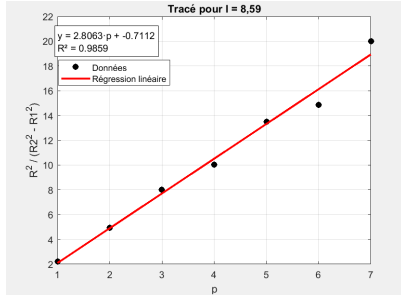
(a) $I = 7.08$ A



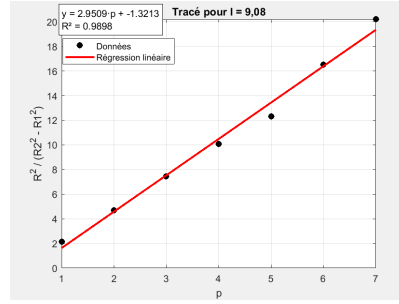
(b) $I = 7.51$ A



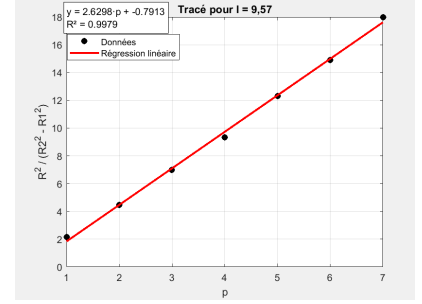
(c) $I = 8.11$ A



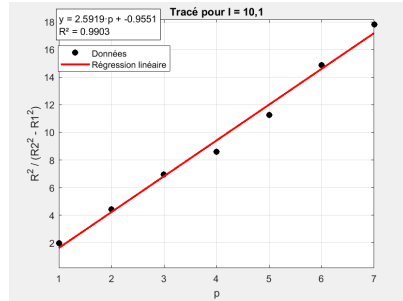
(d) $I = 8.59$ A



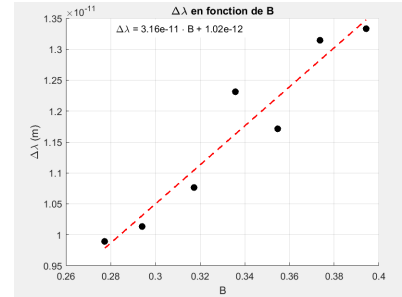
(e) $I = 9.08$ A



(f) $I = 9.57$ A



(g) $I = 10.1$ A



(h) Linear regression of $\Delta\lambda$ as a function of magnetic field.

Figure 5: Determination of the wavelength shift and Bohr magneton from the normal Zeeman effect.

- analyzing only high-quality angular sectors of the image;
- increasing the number of reliable magnetic-field values.

B. Limitations in the Anomalous Zeeman Analysis

For the anomalous Zeeman effect, the number of spectral components is larger, and neighboring interference orders overlap more easily. As a result, assigning each observed ring to a specific transition becomes difficult. An example is shown in Figure 8.

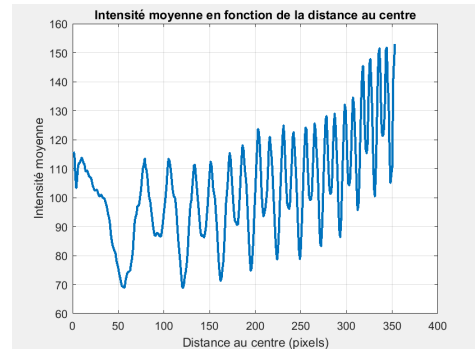
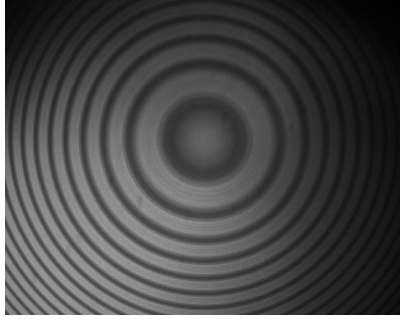
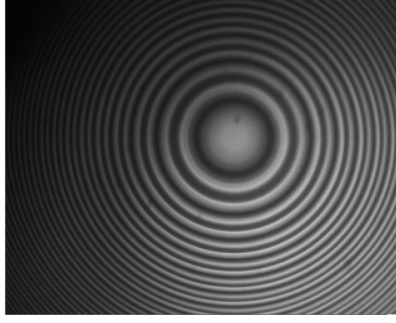


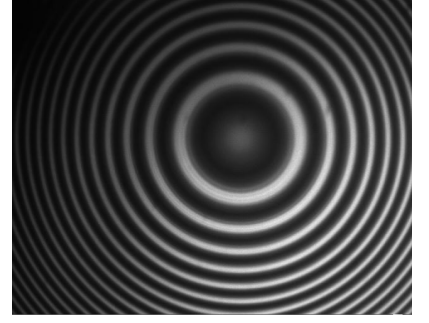
Figure 8: Example of image-analysis limitation in the anomalous Zeeman effect for $I = 8.72$ A. The multiplicity of rings makes sublevel identification difficult.



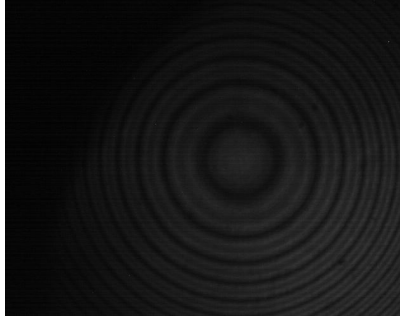
(a) Multiplicity of rings in the anomalous Zeeman effect.



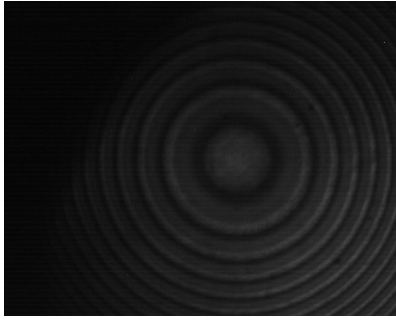
(b) Perpendicular geometry, polarization 0°, $I = 8.12$ A.



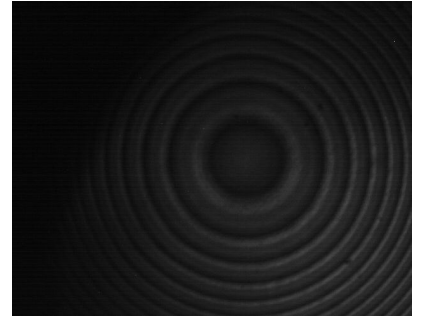
(c) Perpendicular geometry, polarization 90°, $I = 8.12$ A.



(d) Parallel geometry, polarization 0°, $I = 8.12$ A.



(e) Parallel geometry, right circular component selected, $I = 8.12$ A.



(f) Parallel geometry, left circular component selected, $I = 8.12$ A.

Figure 6: Observation of the anomalous Zeeman effect in cadmium. The large number of allowed transitions produces a more complex ring structure than in the normal Zeeman effect.

The anomalous Zeeman observations were therefore mainly qualitative in this report. They nevertheless confirmed the expected increase in spectral complexity compared with the normal Zeeman effect and demonstrated the importance of polarization selection rules.

VII. CONCLUSION

In this experiment, we observed both the normal and anomalous Zeeman effects in cadmium using a Fabry–Pérot interferometer. The normal Zeeman effect produced a splitting into three main components, while the anomalous Zeeman effect produced a more complex ring structure due to the different Landé factors of the two levels involved.

The polarization behavior of the Zeeman components was also verified qualitatively. In the perpendicular observation geometry, the π and σ components exhibited orthogonal linear polarizations. In the parallel geometry, the π component was strongly suppressed, while the $\sigma^\pm$ components remained visible.

From the normal Zeeman effect, we estimated the Bohr magneton. The full data set gave

$$\mu_B = 15.14 \times 10^{-24} \text{ J T}^{-1},$$

while restricting the analysis to the highest-field data gave

$$\mu_B = 11.10 \times 10^{-24} \text{ J T}^{-1}.$$

The latter value is closer to the accepted value

$$\mu_B = 9.274 \times 10^{-24} \text{ J T}^{-1},$$

with a relative deviation of approximately 19.7%. The discrepancy is primarily attributed to uncertainty in ring-radius extraction, optical alignment and limited splitting at low fields.

Overall, the experiment successfully demonstrated how interferometry can resolve very small Zeeman-induced wavelength shifts and how polarization analysis reveals the selection rules governing atomic transitions. It also highlighted the experimental precision required for quantitative atomic spectroscopy.

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